

BUSINESS REQUIREMENTS DOCUMENT

End-to-End Planning & Scheduling Tool

Denim Bottoms and Chinos Manufacturing

Company	Eratex
Document Type	Business Requirements Document
Product Scope	Planning & Scheduling Tool for Garment Manufacturing
Manufacturing Scope	Denim Bottoms + Chinos
Version	1.0
Date	2026-05-26
Status	Draft for Review

1. Executive Summary

Eratex operates a large-scale garment manufacturing environment focused on denim bottoms and chinos. The production flow is complex because it depends on customer approvals, sampling iterations, nominated vendor procurement, fabric quality clearance, planned cut date readiness, sewing line loading, wash complexity, finishing, quality inspection, packing, and shipment commitments.

Although a garment planning tool such as FastReact may already be present in the operating environment, planning activity still appears to remain heavily dependent on manual Excel files, indicating that the formal planning system has not fully become the daily operating backbone of the factory.

Current Business Condition

The factory is meeting customer shipment commitments, but not through a stable and fully trusted planning system. It is maintaining OTIF through manual coordination,

Excel-based firefighting, overtime, expediting, and extra operating expense.

The proposed product is an end-to-end planning and scheduling tool designed to enforce operational discipline across order lifecycle visibility, PCD readiness, weekly planning, daily

production release, workcenter load monitoring, sewing line loading, wash planning, WIP ageing, exception management, and shipment readiness.

The MVP should focus on ten critical operating surfaces that solve the most immediate business problem: converting a complex garment order book into a realistic, executable, constraint-aware, and shipment-protecting production plan.

2. Business Context

2.1 Nature of Eratex Operations

Eratex's manufacturing operation involves large-scale production of denim bottoms and chinos. The production flow is comprehensive:

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Customer enquiry → Sampling → Buyer approval → Order confirmation
→ Fabric and trim procurement → Fabric inward → Fabric QC
→ PCD readiness → Cutting → Sewing → Wet/dry washing
→ Repeat wash or rework if required → Finishing → Final QC
→ Packing → Shipment
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2.2 Operational Sensitivities for Denim and Chinos

- Fabric is often sourced from nominated vendors with lead times of 20–30 days.
- Fabric QC, shade-lot segregation, shrinkage, skewing, and stretch recovery are critical before cutting.
- Washing is a major production stage — not a minor finishing step.
- Wash processes may include dry process, wet wash, enzyme wash, stone wash, bleach, ozone, tinting, softener, and repeat wash cycles.
- Manual effects such as whiskering, scraping, grinding, destroy, and touch-up depend heavily on operator skill.
- Quality failures and rewash loops consume real capacity and affect shipment reliability.
- OTIF can be protected only if the planning system detects risks early and triggers recovery actions in time.

3. Problem Statement

A large denim and chinos manufacturer using FastReact but still relying on Excel is likely operating with a partially digitized planning process where the formal schedule exists in software but real decisions are still made manually.

Eratex is achieving high OTIF, reportedly above 95%, but this performance is being protected through additional operating expense. At the same time, resource utilization is around 70%, which indicates that capacity is not being converted efficiently into stable, predictable, net-good output.

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FastReact or a similar planning tool may be present,
but Excel remains the real execution layer for planning decisions.
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3.1 Operational Gaps Identified

- Planning is formally digital but operationally manual.
- OTIF is maintained through firefighting and cost absorption.
- Utilization is low because flow is unstable.

- Workcenter constraints shift without timely visibility.
- Sewing output may be planned without sufficient wash capacity alignment.
- Wash rework and repeat cycles may not be fully capacity-planned.
- PCD readiness may not be enforced as a hard production gate.
- WIP ageing may remain hidden between departments.
- Daily production release may be driven by urgency rather than readiness.
- Shipment readiness may be assessed too late.

At Eratex scale, small planning gaps multiply across many buyers, styles, production lines, wash routes, quality gates, and shipment commitments. The real problem is not simply the absence of software — the planning system has not become the daily operating discipline of the factory.

4. Business Need

Eratex requires an integrated planning and scheduling tool that can become the single operating layer for production control. The product must help Eratex answer, every day:

Which orders are ready?	Which orders are blocked?
Where is the bottleneck?	What is overloaded?
What can be released today?	What is ageing in WIP?
Which shipment is at risk?	What recovery action is required?
Who owns the next action?	

4.1 Required Transformation

From	To
Manual Excel planning	System-led planning
Department-wise tracking	End-to-end order visibility
Gross capacity planning	Net-good capacity planning
Weekly static planning	Daily executable release
Late-stage firefighting	Early exception detection
OTIF through extra cost	OTIF through stable flow control

5. Business Objectives

5.1 Primary Objectives

1. Establish one integrated planning view from order confirmation to shipment.
2. Enforce PCD readiness before cutting starts.
3. Convert weekly production plans into daily executable releases.
4. Monitor total workcenter load and identify shifting constraints.
5. Improve sewing line loading using realistic capacity assumptions.
6. Treat washing as a core production constraint for denim and chinos.
7. Make WIP ageing visible across departments.
8. Track exceptions with owner, severity, due date, and recovery action.
9. Distinguish production completion from true shipment readiness.

10. Reduce reliance on Excel-based planning and manual firefighting.

5.2 Performance Objectives

Metric	Current Concern	Desired Direction
Resource Utilization	Around 70%	Improve through better readiness, loading, and flow discipline
OTIF	Above 95% but cost-heavy	Maintain or improve while reducing extra operating expense
Overtime Dependency	High due to recovery pressure	Reduce through earlier risk detection
Planning Accuracy	Weak due to Excel dependency	Improve through single system of truth
WIP Ageing	Hidden or manually tracked	Make visible and actionable
Wash Bottleneck Visibility	Often late	Make constraint visible before shipment risk escalates
Rework Impact	May be invisible in capacity	Consume capacity explicitly in plan
Shipment Readiness	Often checked late	Track continuously before dispatch

6. Product Scope

The product scope is divided into two components:

- MVP Scope: Ten critical surfaces required to stabilize planning discipline.
- Mature-State Scope: Additional surfaces required for full-scale deployment and long-term operational maturity.

PART A: MVP SCOPE — TEN CRITICAL SURFACES

7. MVP Surface 1: Order Lifecycle Surface

7.1 Purpose

To provide one digital thread for every customer order from confirmation to shipment.

7.2 Business Problem Solved

Orders are currently tracked across disconnected files and departments. There is no single reliable view of where an order stands, what is pending, who owns the next action, and whether the shipment date is safe.

7.3 Key Capabilities

- Order master view: customer, buyer, style, PO, quantity, delivery date
- Current stage of order with planned vs actual dates by stage
- Pending dependencies and delay reasons
- Owner by stage and risk status
- Next required action
- Linkage to materials, PCD, production, wash, QC, and shipment

7.4 Key Data Elements

Data Element	Description
Order ID	Unique order reference
Customer / Buyer	Customer or brand
Style Code	Garment style
Product Type	Denim bottom, chino, cargo, jogger, etc.
Quantity	Ordered quantity
Delivery Date	Committed shipment date
Current Stage	Current lifecycle status
Risk Status	Green, Yellow, Red, Black
Owner	Responsible user or department
Next Action	Action required to move order forward

7.5 Success Criteria

- Every order has one system-level lifecycle status.
- Users no longer need separate Excel files to know order progress.
- Delayed orders have visible reasons and owners.

8. MVP Surface 2: PCD Readiness Surface

8.1 Purpose

To enforce production readiness before cutting starts.

8.2 Business Problem Solved

Orders may be pushed into cutting before all required conditions are ready, causing downstream rework, holds, shade problems, measurement issues, trim shortages, and shipment risk.

8.3 PCD Validation Checklist

PO confirmed	BOM frozen	Fabric received
Fabric QC passed	Shade lots mapped	Shrinkage report available
Trims available	Pattern approved	Marker ready
PP sample approved	Wash standard approved	Line allocated
Wash capacity booked	QC file ready	

8.4 Readiness Status

Status	Meaning
Ready	Can be released to cutting
Conditionally Ready	Minor open item exists with approved risk
Blocked	Cannot be released
Escalated	Management action required

8.5 Success Criteria

- No order is cut without readiness visibility.
- Blocked orders are visible before PCD.
- Conditional releases are approved and auditable.

9. MVP Surface 3: Weekly Planning Workbench

9.1 Purpose

To convert the order book into an executable weekly cross-functional plan.

9.2 Business Problem Solved

Weekly plans often fail because material readiness, cutting, sewing, wash, finishing, QC, and shipment are not planned as one integrated flow.

9.3 Key Capabilities

- 2–4 week firm plan view with 8–12 week rolling visibility
- Order sequencing and cutting plan
- Sewing line allocation and wash plan
- Finishing, inspection, and shipment plan
- Load vs capacity view with material readiness filter
- Shipment-risk priority indicator
- Plan freeze capability and plan-change impact visibility

9.4 Success Criteria

- Weekly plan becomes the coordination contract across departments.
- Plan is capacity-checked before being frozen.
- Orders with material or PCD risks are not silently included in execution plan.

10. MVP Surface 4: Daily Production Release Surface

10.1 Purpose

To release only what is actually ready and executable today.

10.2 Business Problem Solved

Daily production may be released based on urgency rather than actual readiness, resulting in line stoppages, partial production, WIP imbalance, and manual firefighting.

10.3 Daily Release Validation Gates

Material availability	Previous process completion
Machine availability	Manpower availability
Quality clearance	Workcenter capacity
Next-process readiness	

10.4 Release Actions Supported

- Release to cutting, sewing, wash, or finishing
- Release block with reason and exception approval
- Daily release history and audit trail

10.5 Success Criteria

- Daily release becomes gate-based.
- Unready orders are blocked or escalated.
- Production teams receive executable work, not theoretical plan items.

11. MVP Surface 5: Workcenter Load Monitor

11.1 Purpose

To show load, capacity, queue, WIP ageing, utilization, and current constraint across workcenters.

11.2 Business Problem Solved

The active bottleneck can shift from sewing to washing, dry process, finishing, QC, or shipment — but this is often discovered late.

11.3 Workcenters in MVP Scope

Workcenter
Fabric QC
Cutting
Sewing Lines
Dry Process
Wet Wash
Drying
Finishing

Workcenter

Packing

Final QC

Shipment Documentation

11.4 Key Metrics

Metric	Description
Available Capacity	Capacity by day or week
Planned Load	Work already scheduled
Queue	Work waiting before workcenter
WIP Ageing	Time waiting before next process
Utilization	Load as percentage of available capacity
Throughput	Actual output
Constraint Flag	Whether workcenter is bottleneck
Recovery Need	Whether action is required

11.5 Success Criteria

- Bottleneck shifts are visible early.
- Planners can see total load, not just urgent orders.
- Recovery action is triggered before shipment risk becomes critical.

12. MVP Surface 6: Sewing Line Loading Surface**12.1 Purpose**

To support realistic sewing line allocation and production target setting.

12.2 Business Problem Solved

Sewing lines are often loaded using nominal capacity, while real output depends on style SMV, operator skill, absenteeism, learning curve, quality loss, line balance, and machine availability.

12.3 Key Capabilities

- Assign orders to lines and calculate daily target from SMV and manpower
- Track line capacity by day and loading by style and order
- Capture actual output and compare planned vs actual
- Calculate gross output and net-good output with quality loss tracking
- Identify overloaded or underloaded lines
- Suggest alternate line or split-line loading

12.4 Required Inputs

Style SMV	Operation bulletin	Line manpower
Machine availability	Operator skill level	Target efficiency

12.5 Success Criteria

- Sewing plan reflects realistic net-good capacity.
- Underperformance is visible by line and order.
- Line loading decisions improve resource utilization.

13. MVP Surface 7: Wash Planning Surface

13.1 Purpose

To plan washing as a core production constraint, not as an afterthought.

13.2 Business Problem Solved

In denim manufacturing, wash processes and rewash loops can become the real bottleneck. If washing is planned only after sewing, WIP builds up and shipment risk increases.

13.3 Key Capabilities

- Define wash route by style and plan dry and wet process
- Plan washer/dryer capacity and create wash batches
- Maintain shade-lot identity throughout the wash process
- Track batch queue and wash completion
- Track rewash or touch-up requirements
- Reserve capacity for repeat wash and link wash status to finishing release

13.4 Wash Routes Covered

Dry process	Wet wash	Enzyme wash
Stone wash	Bleach / Ozone	Tinting
Softener	Hydro extraction	Drying
Post-wash shade check	Rewash / touch-up loop	

13.5 Success Criteria

- Sewn garments do not pile up invisibly before wash.
- Wash bottleneck is visible before shipment risk escalates.
- Rewash consumes planned capacity.

14. MVP Surface 8: WIP and Queue Monitoring Surface

14.1 Purpose

To monitor work waiting between processes and surface hidden delays.

14.2 Business Problem Solved

Hidden WIP causes hidden delay. Departments may report output, but orders may remain stuck between processes.

14.3 WIP Points Tracked

Fabric waiting for QC	Fabric cleared but not cut
Cut panels waiting for sewing	Sewn garments waiting for wash
Washed garments waiting finishing	Finished garments waiting packing
Packed cartons waiting inspection	

14.4 Key Capabilities

Capability	Description
WIP Quantity	Quantity by stage
Ageing by Stage	Time at current stage
Hold Reason	Why the WIP is held
Next Process	Where it should flow
Owner	Responsible person
Shipment Risk	Risk level for linked order
Ageing Threshold	Escalation trigger point

14.5 Success Criteria

- Hidden queues become visible.
- Ageing WIP is escalated before it becomes a shipment risk.
- Planners can prioritize flow, not only output.

15. MVP Surface 9: Exception and Alert Surface

15.1 Purpose

To drive exception-based planning and structured recovery action.

15.2 Business Problem Solved

Operational issues are often known informally but not structured into a system with severity, owner, due date, and recovery action.

15.3 Exception Types and Examples

Fabric delayed beyond PCD	Fabric QC failed
PP sample approval pending	Line overloaded next week
Wash queue exceeds capacity	Sewn WIP ageing above limit
Post-wash rejection high	Final inspection not scheduled
Shipment risk turned red	Recovery action overdue

15.4 Exception Data Structure

Field	Description
Exception Type	Material, capacity, quality, approval, WIP, shipment
Affected Order	Order impacted by the exception
Severity	Green, Yellow, Red, Black or equivalent

Field	Description
Owner	Responsible person
Due Date	Resolution target
Suggested Action	Recommended recovery
Status	Open, In Progress, Closed
Escalation Level	Normal, Urgent, Management

15.5 Success Criteria

- Exceptions are not buried in WhatsApp or Excel.
- Every critical issue has an owner and due date.
- Management sees only actionable risks.

16. MVP Surface 10: Shipment Readiness Surface

16.1 Purpose

To protect final shipment commitments and provide continuous dispatch-readiness visibility.

16.2 Business Problem Solved

Production completion does not automatically mean shipment readiness. Orders may be delayed by final QC, buyer inspection, packing, labels, documents, or forwarder booking.

16.3 Shipment Readiness Checklist

Final QC passed	AQL passed	Packing complete
Cartons closed	Barcode/label correct	Packing list ready
Invoice ready	Forwarder booked	Shipment date confirmed

16.4 Key Tracking Capabilities

- Shipment date, finished quantity, packed quantity, and short quantity
- Final QC, AQL / buyer inspection, and carton readiness status
- Barcode / label readiness and documentation status
- Shipment booking and split shipment decisions

16.5 Success Criteria

- Shipment risk is visible before dispatch date.
- Production completion and dispatch readiness are clearly separated.
- OTIF is protected with fewer last-minute interventions.

PART B: MATURE-STATE SCOPE — ADDITIONAL SURFACES

The MVP stabilizes planning discipline. A mature deployment extends the product into the following additional surfaces:

#	Surface	Purpose
17	Executive Control Tower	Management-level exception dashboard with risk heatmap, OTIF cost, overtime exposure, and customer-wise delay view
18	Enquiry and Costing	Pre-order feasibility evaluation including capacity simulation, wash complexity assessment, and delivery promise validation
19	Sampling and Approval Tracker	Proto, fit, wash, size set, and PP sample workflow with buyer comments, resubmission tracking, and approval ageing
20	Style Master and Technical File	Style-level technical master including tech pack, BOM, SMV, operation bulletin, machine/skill requirements, and wash route
21	BOM and Material Planning	Material requirement calculation with shortage alerts, fabric width and shrinkage impact, and procurement linkage
22	Procurement and Vendor Follow-up	Vendor PO tracking, ETA monitoring, delay alerts, material criticality ranking, and vendor performance scorecard
23	Fabric Inward and Fabric QC	Roll-wise receipt and 4-point inspection including shade lot, GSM, shrinkage, skewing, and stretch/recovery testing
24	Operator Skill and Capacity	Operator skill matrix with efficiency, quality rating, absenteeism tracking, learning curve, and training needs
25	Cutting Room	Shade-wise cutting management including marker approval, spreading, cut panel QC, bundle numbering, and sewing issue
26	Wash Recipe and Batch Execution	Shopfloor wash batch control with recipe steps, machine assignment, shade/hand-feel results, and rewash management
27	Quality Management	Full QC coverage from fabric through final inspection with defect categorization, root cause, and quality-adjusted capacity
28	Rework and Recovery	Structured rework order management with type, quantity, capacity reservation, shipment impact, and closure tracking
29	Calendar / Gantt Planning	Visual order, department, line, wash, shipment, approval, and procurement timelines

#	Surface	Purpose
30	Recovery Planning / What-If Simulation	Scenario testing for fabric delays, line splits, wash rework, overtime, and shift extensions with cost and capacity impact
31	Plan Change and Approval	Governance workflow for frozen/firm zone plan changes with reason codes, impact analysis, and audit trail
32	Department Handover	Formal interdepartmental handover surface with completeness checks across all production stages
33	Master Data Governance	Completeness and version control for styles, BOM, SMV, workcenters, shift calendars, and wash recipe masters
34	Performance Analytics	OTIF, utilization, plan adherence, line efficiency, net-good output, rework rate, and cost of recovery trends
35	Mobile / Shopfloor Update	Low-friction capture for output, defects, shortages, machine issues, absenteeism, and handover confirmations
36	Role-Based Home Surfaces	Role-specific operating dashboards for all user functions from merchandiser to management

PART C: FUNCTIONAL REQUIREMENTS

37. Core Functional Requirements

Req. ID	Requirement
FR-001	System shall maintain order lifecycle status from confirmation to shipment.
FR-002	System shall maintain planned and actual dates for each major order stage.
FR-003	System shall enforce PCD readiness checks before cutting release.
FR-004	System shall support weekly capacity-based production planning.
FR-005	System shall support daily production release based on readiness.
FR-006	System shall monitor load and capacity at each workcenter.
FR-007	System shall identify current and future bottlenecks.
FR-008	System shall support sewing line loading using SMV, manpower, and capacity assumptions.
FR-009	System shall distinguish gross output from net-good output.
FR-010	System shall support wash route and wash batch planning.
FR-011	System shall track rewash and repeat wash capacity consumption.
FR-012	System shall track WIP quantity and ageing between processes.

Req. ID	Requirement
FR-013	System shall generate exceptions for material, capacity, quality, approval, WIP, and shipment risks.
FR-014	System shall assign owner and due date to every exception.
FR-015	System shall track shipment readiness separately from production completion.
FR-016	System shall maintain audit trail for plan changes and conditional releases.
FR-017	System shall support role-based views and permissions.
FR-018	System shall support integration with ERP or existing order/material data sources.
FR-019	System shall support export and reporting for management review.
FR-020	System shall preserve historical data for performance analytics.

38. Key Business Rules

38.1 PCD Release Rules

- No order should move to cutting without PCD readiness status.
- Blocked orders must have open dependency and owner.
- Conditional release must require reason and approval.
- Fabric QC failure should block PCD unless overridden by authorized role.

38.2 Planning Rules

- Weekly plan must be capacity checked before freeze.
- Frozen-zone changes must require reason code and approval.
- Daily release must validate real readiness.
- Workcenter overload should trigger exception.
- Orders with high shipment risk should receive priority review.

38.3 Capacity Rules

- Capacity should be based on available minutes, manpower, machine availability, SMV, and demonstrated efficiency.
- Net-good output should account for quality loss and rework.
- Rework and rewash must consume capacity.
- Wash capacity must be separately planned from sewing capacity.

38.4 WIP Rules

- WIP must be tracked at key handover points.
- WIP ageing beyond threshold must trigger alert.
- WIP with quality hold must not proceed without clearance.
- WIP waiting before constraint workcenter must be prioritized by shipment risk and readiness.

38.5 Shipment Rules

- Shipment readiness must include final QC, inspection, packing, documents, and booking.
- Production complete does not mean shipment ready.
- Short shipment risk must be visible before shipment date.
- Split shipment decision must be tracked and approved.

PART D: NON-FUNCTIONAL REQUIREMENTS

39. Non-Functional Requirements

Category	Requirement
Usability	Screens must be simple, action-oriented, and suitable for planners and shopfloor supervisors.
Performance	Workcenter load, WIP, and order risk views should load quickly for large order volumes.
Scalability	System must support large factory operations with many lines, workcenters, styles, and orders.
Reliability	Planning data must be version-controlled and auditable.
Security	Access should be role-based. Sensitive commercial and customer data must be protected.
Integration	System should integrate with ERP, material records, shopfloor actuals, and shipment data where available.
Auditability	Plan changes, overrides, conditional releases, and shipment changes must be traceable.
Configurability	Workcenters, statuses, thresholds, calendars, and roles must be configurable.
Mobile Readiness	Shopfloor update flows should eventually support mobile or tablet usage.
Data Quality	Missing or incomplete master data must be visible before planning decisions.

PART E: INTEGRATION REQUIREMENTS

40. Required Integration Areas

Integration Area	Purpose
Order Management / ERP	Order, customer, style, PO, quantity, delivery date
BOM / Material System	Fabric, trims, packing materials
Procurement	Vendor PO, ETA, inward status
Fabric QC	Inspection status and results
Production Actuals	Cutting, sewing, wash, finishing output
Quality System	Defects, holds, rework, AQL
HR / Attendance	Operator availability and absenteeism
Machine / Maintenance	Machine availability and breakdown
Shipment / Logistics	Packing, inspection, dispatch, documents

PART F: ROLES AND USERS

41. Key User Roles

Role	Key Usage
Management	Control tower, shipment risk, bottlenecks, performance
Production Planner	Weekly plan, daily release, workcenter load, recovery
Merchandiser	Order lifecycle, approvals, buyer comments
Procurement User	Material status, vendor follow-up
Fabric QC User	Fabric inspection and clearance
Cutting Manager	Cutting readiness and cut output
Sewing Manager	Line load, output, WIP, constraints
Washing Manager	Wash plan, batch status, rewash
Finishing Manager	Finishing queue, packing readiness
QC Manager	Quality holds, inspection, defects
Shipment Team	Shipment readiness and dispatch

Role	Key Usage
Shopfloor Supervisor	Daily updates, output, issues, handover

PART G: REPORTING REQUIREMENTS

42. MVP Reports

Report	Purpose
Order Status Report	End-to-end order progress
PCD Readiness Report	Orders ready, blocked, conditional
Weekly Plan Report	Department-wise weekly plan
Daily Release Report	Work released and blocked
Workcenter Load Report	Load vs capacity by workcenter
Sewing Line Load Report	Line-wise plan and output
Wash Queue Report	Wash load, queue, and rewash
WIP Ageing Report	Stuck WIP and owner
Exception Report	Open issues and recovery actions
Shipment Readiness Report	Orders ready or at risk for shipment

PART H: SUCCESS METRICS

43. Business Success Metrics

Metric	Target Direction
OTIF	Maintain above 95% while reducing recovery cost
Utilization	Improve from 70% through better flow and readiness
Overtime Cost	Reduce overtime used for shipment recovery
Planning Adherence	Improve weekly and daily plan compliance
PCD Readiness Accuracy	Reduce cutting releases with open blockers
WIP Ageing	Reduce ageing at key handover points

Metric	Target Direction
Wash Queue Ageing	Reduce sewn garments waiting for wash
Rework Visibility	100% of major rework captured with owner
Shipment Readiness	Improve advance visibility of dispatch blockers
Excel Dependency	Reduce manual planning sheets over time

PART I: IMPLEMENTATION ROADMAP

44. Suggested Implementation Phases

Phase 1: MVP Foundation

Build and deploy the ten critical surfaces:

- | | |
|-----------------------------------|-----------------------------|
| 1. Order Lifecycle | 2. PCD Readiness |
| 3. Weekly Planning Workbench | 4. Daily Production Release |
| 5. Workcenter Load Monitor | 6. Sewing Line Loading |
| 7. Wash Planning | 8. WIP and Queue Monitoring |
| 9. Exception and Alert Management | 10. Shipment Readiness |

Phase 2: Material, Sampling, and Technical Master Expansion

Add the following capabilities:

- | | |
|----------------------------------|---------------------------|
| Enquiry and costing | Sampling and approvals |
| Style technical file | BOM and material planning |
| Procurement and vendor follow-up | Fabric inward and QC |

Phase 3: Shopfloor and Quality Depth

Add the following capabilities:

- | | |
|-----------------------------|-----------------------|
| Cutting room | Wash recipe execution |
| Quality management | Rework and recovery |
| Operator skill and capacity | Department handover |
| Mobile shopfloor updates | |

Phase 4: Governance, Simulation, and Analytics

Add the following capabilities:

- | | |
|--------------------------|---------------------------|
| Executive control tower | Calendar / Gantt planning |
| What-if simulation | Plan change approval |
| Master data governance | Performance analytics |
| Role-based home surfaces | |

PART J: KEY RISKS AND MITIGATION

45. Implementation Risks

Risk	Mitigation
Users continue using Excel	Make MVP surfaces action-critical and reduce duplicate entry
Poor master data	Start with minimum required master data and completeness checks
Delayed shopfloor actuals	Use simple update flows and supervisor-level capture
Resistance to gate-based release	Use management-backed policy for PCD and daily release
FastReact overlap confusion	Define integration or coexistence role clearly
Wash complexity under-modeled	Include wash planning in MVP, not later phase
Data not trusted	Show planned vs actual and audit trail
Too many alerts	Prioritize shipment-impacting exceptions
Large-scale rollout risk	Pilot with selected lines/styles before scale-up
Extra workload perception	Replace Excel routines instead of adding parallel work

PART K: OPEN DECISIONS

46. Decisions Required Before Build

11. Will this tool replace FastReact, integrate with it, or operate as an operational control layer above/beside it?
12. What is the source of truth for order master and PO details?
13. What is the source of truth for material readiness?
14. What production actuals are currently captured digitally?
15. Which workcenters should be included in the first pilot?
16. What are the current PCD readiness rules at Eratex?
17. How is wash planning currently performed?
18. How are sewing line capacities currently calculated?
19. How is overtime cost currently tracked against shipment recovery?
20. Which Excel files are currently used for real planning decisions?

47. Final BRD Summary

Eratex requires an end-to-end planning and scheduling tool because its current planning environment appears to be partially digitized but still operationally dependent on Excel. The factory maintains high OTIF, but with additional operating expense, while resource utilization remains around 70%. This suggests that shipment commitments are being protected through late-stage recovery rather than stable flow-based planning.

The proposed tool must become the daily operating discipline layer for Eratex. It must enforce order visibility, PCD readiness, weekly and daily planning control, workcenter load monitoring, realistic sewing line loading, denim-specific wash planning, WIP ageing visibility, exception ownership, and shipment readiness.

The MVP must deliver the ten critical surfaces needed to stabilize the planning process. The mature-state product should then expand into sampling, procurement, quality, shopfloor execution, skill-based capacity, what-if simulation, master data governance, analytics, and role-based control surfaces.

Intended Business Outcome

Maintain high OTIF,
Increase resource utilization,
Reduce overtime and recovery cost,
Reduce Excel dependency,
Improve planning discipline,
and protect shipment commitments through early, constraint-aware control.